

Rupanshu Shah

PhD Scholar · Computer Science and Engineering · IIT Delhi

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RESEARCH INTERESTS

Quantum Algorithms • Fine-grained Complexity • Quantum Walks • Complexity Theory • Cryptography • CS Theory

EDUCATION

Ph.D., Computer Science and Engineering, IIT Delhi — Jul 2025–present

Advisor: Prof. Rajendra Kumar • CGPA 9.313 • Institute Assistantship (also qualified for JRF)

B.Tech., Computer Science and Engineering, IIT Delhi — 2023

CGPA 7.2/10 • Department Specialization in Theoretical Computer Science: 4 additional Master's/PhD-level courses and a second major project (Quantum ML), beyond the B.Tech requirements

RESEARCH EXPERIENCE

PhD Research, IIT Delhi (Advisor: Prof. Rajendra Kumar)

Jul 2025–present

- Spent two semesters building a toolkit of quantum algorithmic techniques (amplitude amplification and estimation; classical and quantum walks, including finding marked vertices in arbitrary graphs; quantum divide and conquer, multidimensional quantum walks, transducers) and lower-bound techniques (adversary method, polynomial method, Zhandry's compressed oracle).
- Studying problems in P through a quantum lens: optimal quantum algorithms and matching quantum lower bounds.
- Current focus: quantum fine-grained complexity of All-Pairs Shortest Paths (APSP): quantum analogues of the APSP and Negative-Weight Triangle conjectures, their tightness, and quantum variants of APSP.

Predocutorial Fellow, IIT Delhi (Prof. Rajendra Kumar)

Aug 2024–Jul 2025

- Surveyed results and techniques from 4+ research papers on instance compression.
- Studied k-SAT and analysed a quantum approach yielding new quantum poly-time compression bounds.
- Investigated classical and quantum methods for Set Cover compression.
- Explored oracle communication protocols for instance compression in the quantum setting.

INTERNSHIP

ISRO – Indian Space Research Organization

Jun 2021–Jul 2021

- Developed Python software for processing Chandrayaan-2 SAR images.
- Aligned and stitched 1000+ DFSAR images with typical image size in GBs.
- Tested 5 interpolation techniques; finalized 2D cubic convolution.
- Project deployed in determining the Chandrayaan-3 landing site on the lunar surface.

MANUSCRIPTS AND NOTES

- Analyzing Complexity Class P through a Quantum Lens: Fine-Grained Bounds and APSP. Working draft for PhD comprehensive exam, Oct 2026.
- Classical and Quantum Random Walks. Notes from classical random walks and spectral gaps to quantum walks, glued trees and spatial search.
- Label Cover to CVP Reduction.
- A Proposed Conjecture on Projection Games.

TALKS AND CONFERENCES

- “From Grover Search to Quantum Walks: Search under Structure”, CSE Department Symposium, IIT Delhi, May 2026.
- Attended FSTTCS 2025, BITS Pilani Goa Campus, Dec 2025.

TEACHING (TEACHING ASSISTANT, IIT DELHI)

COL759 Cryptography — Sem I 2026–27

COL1000 Introduction to Computer Science — Summer 2026

COL226 Programming Languages — Sem II 2025–26

COL106 Data Structures and Algorithms — Sem I 2025–26

PROJECTS

Quantum Finite Automata & Quantum Languages (Jan 2023–May 2023)

- Independent study: analysed 3 research papers and 1 literature survey on quantum finite automata.
- Identified a potentially tractable research problem and presented it at the CS Theory Seminar.

Maximum Independent Set in Degree-3 Graphs (Dec 2021–May 2022)

- Rigorously analysed 60+ cases of the $O(1.0821^n)$ algorithm for bounded degree-3 graphs and found a critical bug.

Other Projects

- Designed PageRank from scratch using the MapReduce framework on MPI and OpenMP.
- Implemented Bleichenbacher's attack on RSA PKCS#1 v1.5 encryption.
- Devised and implemented (Python) an algorithm for renaming CNF to Horn formulas.
- Performed a buffer overflow attack on the xv6 OS.

SCHOLASTIC ACHIEVEMENTS

- UGC NET (Computer Science), Dec 2024: AIR 4; qualified for Junior Research Fellowship (JRF).
- GATE CS 2025: score 831, AIR 221.
- GRE 337/340 · TOEFL 115/120 · JEE Main 99.72 percentile; 1 of 120 students selected from 1 million candidates to study Computer Science at IIT Delhi.
- NTSE Scholar: top 0.125% of 800,000 candidates; annual scholarship of INR 24,000 from the Government of India.
- KVPY Scholar: top 1%, one of 1,000 scholars from 100,000+ candidates across India.
- UPSTSE Scholar: INR 24,000 annually, Council of Science & Technology, Government of U.P.
- Gold Medal, Individual Contest, International Young Mathematicians Convention (IYMC) 2017: 1000+ participants from 15+ countries.
- State Rank 7, Regional Mathematical Olympiad (RMO) 2017, HBCSE.
- Qualified NSEA'17 (Astronomy) and NSEC'19 (Chemistry), IAPT: 1 of 300 across India.

SKILLS

Programming: C, C++, Python, Java, SQL, SAS · Frameworks: Qiskit, OpenMP, MPI, LaTeX · Languages: German, French

RELEVANT COURSES

Theory of Computing, Logic, Quantum Computing, Randomized Algorithms, Cryptography, Complexity Theory, Blockchain & Smart Contracts, Formal Verification, Lean Programming

CERTIFICATIONS

AWS Machine Learning · Google Cloud Machine Learning · Linear Regression (Coursera) · Blockchain Specialization (Coursera, 3 courses) · SAS Programming (SAS Institute)

EXTRACURRICULARS

- Initiated and managed a Dog Feeding Club at my hostel: fund collection, medication and food for resident dogs.
- Taught underprivileged children under Project Each One Teach One.
- Created teaching material for underprivileged students under the National Service Scheme (NSS).
- 4th position among players from 13 hostels in the Inter-hostel Chess Competition.